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Mechanical and structural properties of phosphate glasses

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Abstract

Mechanical and structural properties of sodium (NAFP) and zinc (ZAFP) iron–aluminum–phosphate bulk glass and fibers have been investigated. Young's modulus of the fibers was measured by a three-point bending method while the strength was measured by a two-point bending method. In general, the tensile strength of the ZAFP fibers (4.2–7.2 GPa) was higher than the tensile strength of the NAFP fibers (2.8–4.2 GPa). After exposing the fibers to air for 10 days, the strength decreased by 15–34%. The structure of bulk glass as well as fibers, studied by Mössbauer and IR spectroscopy, was very similar for all the compositions studied. © 2001 Elsevier Science B.V. All rights reserved.

1. Introduction

The practical application of phosphate glasses is often limited by their poor chemical durability. Recently, several phosphate glasses with high aqueous corrosion resistance have been reported [1]. Their properties such as low melting point, high thermal expansion coefficient, and optical properties make these glasses potential candidates for many technological applications such as sealing materials, medical use, and solid state electrolytes [1]. Because of their unusually high chemical durability and low processing temperature, iron phosphate glasses are being considered for vitrifying certain nuclear wastes that are poorly suited for borosilicate glasses [2,3].

Phosphate glasses are also becoming important in optical technology such as in high energy laser applications [1]. Because of the mechanical and thermal stresses that they are subjected to in some of these applications, knowledge of their strength is important. Even though the structure and properties of bulk iron phosphate based glasses have been studied [4–8] previously, there is little information for the mechanical properties of iron phosphate fibers compared to commercial silicate fibers. The technological applications of these glasses are clear, so a better understanding of the mechanical properties of phosphate glasses could lead to further applications or improved glass compositions.

In the present study, we have investigated mechanical properties such as Young's modulus and tensile strength, and structural features of sodium

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(NAFP) and zinc (ZAFP) aluminum–iron–phosphate glass fibers and corresponding bulk glasses. The structure of the bulk glass and the fibers was studied by Mössbauer and IR spetctroscopy

2. Experimental procedures

2.1. Preparation of glass and fibers

Glasses were prepared by melting homogeneous mixtures of reagent grade chemicals in high purity alumina crucibles at temperatures between 1000°C and 1200°C for approximately 2 h in air. The batch compositions of the glasses are given in Table 1. Glasses in the first series had the general composition of $20Na_2O$–$(20-x)Al_2O_3$–xFe_2O_3–$60P_2O_5$ (where $x = 0, 5, 10, 15, 20$) and are labeled as NAFP glasses in Table 1. Glasses in the second series had a general composition of 15ZnO–$(17.5-x)Al_2O_3$–xFe_2O_3–$67.5P_2O_5$ (where $x = 0$, $7.5, 12.5, 17.5$) and are labeled as ZAFP glasses in Table 1. Bulk glasses were obtained by pouring the melt into a steel mould to form rectangular bars with the approximate dimensions of $1 \times 1 \times 5$ cm^3 which were annealed at 475°C for ~1 h and slowly cooled to room temperature. Powders of the annealed glasses were checked for crystalline phases by X-ray diffraction and no crystalline phase was detected. Fibers with a diameter between 50 and 200 μm and approximately 50 cm in length were drawn by hand from the melt in a crucible using an alumina bait rod.

2.2. Young's modulus and strength measurements

A three-point bending test was used to measure Young's modulus of the fibers at room temperature [9,10]. The span of the bending test set-up was 30 mm in order to have $L \gg d$ where L is the span width and d is the fiber diameter, between 50 and 140 μm. In the experimental setup, the middle of the fiber was fixed while the ends were bent at a controlled speed of 0.05 mm/s. The deflection was automatically measured by the Instron instrument while the applied load was measured by a digital balance. Young's modulus, E, at room temperature, was calculated using the equation from beam deflection theory for three-point bending,

$$E = L^3P/48\delta I, \quad (1)$$

where P is the load applied, δ is the deflection, and I is the moment of inertia about the neutral axis for cylindrical samples ($\pi d^4/4$). The slope (P/δ) was computed by a least-squares fitting of the linear part (elastic region) of the load/deflection curve. Measurements were made on a minimum of five fibers for each composition and the average modulus values are given in Table 2. The estimated error in E was ±15%.

In general, the tensile strength of a fiber is measured by pulling a length of a fiber in tension until it fails. In the present study, a two-point bending technique was used [11]. In this method, a fiber is carefully bent and placed between two ground and polished faceplates until it is U-shaped

Table 1
Batch compositions and selected properties of bulk sodium (zinc)–aluminum–iron–phosphate glasses

Sample	Batch composition (mol%)					Glass density (g/cm^3)	Glass transition temperature, T_g (°C)	Glass softening temperature, T_s (°C)	Thermal expansion coefficient, α ($\times 10^{-7}$/°C)	Dissolution rate (8 days at 90°C) (g/cm^2 min)
	Na_2O	ZnO	Al_2O_3	Fe_2O_3	P_2O_5					
NAP3	20	0	20	0	60	2.61	480	530	122	3.4×10^{-8}
NAFP1	20	0	15	5	60	2.69	504	554	106	3.1×10^{-8}
NAFP2	20	0	10	10	60	2.75	504	548	101	1.0×10^{-7}
NAFP3	20	0	5	15	60	2.79	515	558	101	1.7×10^{-7}
NFP3	20	0	0	20	60	2.84	484	536	108	3.5×10^{-7}
ZAP1	0	15	17.5	0	67.5	2.68	629	687	70	6.4×10^{-8}
ZAFP2	0	15	10	7.5	67.5	2.79	546	601	66	4.5×10^{-7}
ZAFP1	0	15	5	12.5	67.5	2.81	474	520	70	2.1×10^{-7}
ZFP1	0	15	0	17.5	67.5	3.39	478	532	69	1.6×10^{-8}
ZFP2	0	15	0	25	60	3.14	470	515	64	3.4×10^{-9}
ZAP2	0	15	25	0	60	2.63	659	714	67	1.1×10^{-8}

Table 2
Young's modulus (E) and tensile strength of glass fibers

Fiber	E (GPa)[a]	σ (GPa)[b]
SiO_2[c]	70	14
E glass[c]	70	5.8
S-Glass[c]	87	8.4
$NaPO_3$[c]	36	1.8
$Zn(PO_3)_2$[c]	42	2.4
F40	69	7.2
NAP3	60	2.8
NAFP1	67	3.7
NAFP2	67	4.2
NAFP3	68	3.6
NFP3	69	2.8
ZAP1	67	6.2
ZAFP2	66	4.7
ZAFP1	72	5.4
ZFP1	68	4.3
ZFP2	72	4.8
ZAP2	89	7.2

The results are also compared to fused silica fiber and other commercial fibers reported in the literature.
[a] Average of 5–7 fibers, rounded to nearest 1 GPa.
[b] Tensile strength at 50% failure probability.
[c] These values are taken from [15].

(see Fig. 1) [11]. The ends are clamped and the face plates are then driven together by a computer-controlled stepper motor until fracture of the bent fiber is sensed by an acoustic detector. The tensile stress at failure in the bent part of the fiber (which is untouched) is calculated from the tensile strain measured from the two-point bending test using the equation [11]

$$\sigma = 1.198Ed/(D - d), \tag{2}$$

where E is the room temperature Young's modulus, D is the separation of the face plates at failure,

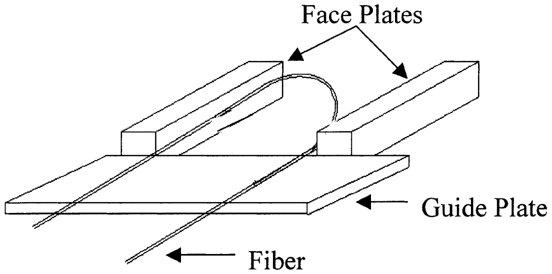

Fig. 1. Schematic illustration of the bending device used for the tensile strength measurements.

and d is the fiber diameter. One of the advantages of this bending technique, over the pure tensile method, is that only a small length of the fiber is under high stress in this test so the fiber diameter can be measured close to the fracture. The derivation of Eq. (2) and a comparison of tensile and bending methods are given in [11,12]. To eliminate any static fatigue or delayed failure, all measurements were made in liquid nitrogen. This was achieved by immersing the fiber loaded apparatus into a liquid nitrogen dewer. Tests were conducted on both freshly pulled fibers (within 10–120 min of being pulled) and on fibers exposed to air for several days. The fibers used for measuring Young's modulus and the bending strength were uncoated. Care was taken so that the small section of the fiber being tested was not touched in any way or by any material (other than air) prior to testing.

Fibers ~5 cm long were tested and these short sections had a variation in diameter of ~8% which will result in a variation in strength values. The cumulative probability of failure by an applied stress, σ_{max}, is given by

$$F(\sigma_{max}) = 1 - \exp\left[-\int_A f(\sigma)\mathrm{d}A\right], \tag{3}$$

where $f(\sigma)$ is interpreted as the probability per unit area that failure has occurred by a stress σ. The Weibull distribution function is commonly used to represent $f(\sigma)$ [11]. These variations in strength will yield a Weibull modulus.

2.3. Mössbauer and IR measurements

It is known that a binary Fe_2O_3–P_2O_5 starting composition changes to a ternary glass composition during melting since some of the Fe(III) ions in the batch (Fe_2O_3) are reduced to Fe(II) ions [4–8]. Mössbauer spectroscopy was used to determine the iron valence and to explore structural changes in the iron environment between bulk glass and corresponding fibers. The measurements were conducted at room temperature using an ASA 600 spectrometer which utilized a 35 mC ^{57}Co source embedded in a rhodium matrix. The spectrometer was calibrated with a metallic α-iron foil whose line width was 0.26 mm/s. Each Mössbauer spectrum was fitted with eight broadened paramagnetic Lorentzian doublets which yields reliable average hyperfine parameters [4,5,13].

The IR spectra were collected between 400 and 2000 cm^{-1} using an FTIR spectrometer. Samples were prepared as pellets by pressing a mixture of 4 mg of glass powder and 150 mg of anhydrous KBr powder. 20 scans were collected for each sample and averaged. The spectrum for pure KBr was subtracted from each glass spectrum to correct the background.

2.4. Other properties

The chemical durability of bulk glass samples approximately $1 \times 1 \times 1$ cm^3 in size was evaluated from the dissolution rate (DR) in distilled water at 90°C. The DR was calculated from the measured weight loss, ΔW (g), sample surface area A (cm^2) and the immersion time t (min), using the equation $DR = \Delta W/At$. All measurements were made in duplicate and the average value is reported herein. The estimated error in DR is ±15%.

The density of each glass was measured by Archimede's method using distilled water as the buoyancy liquid. The average thermal expansion coefficient (α) from 100°C to 300°C and the dilatometric transition temperature (T_g) were measured using a dilatometer which was calibrated using a standard silver rod. The measurements were made using samples 2.54 cm in length heated at a rate of 5°C/min in air. The estimated error in density, α, and T_g is ± 0.02 g/cm^3, $\pm 0.05 \times 10^{-7}$/°C, and ±5°C, respectively.

3. Results

3.1. Glass properties

The batch composition and selected properties are listed in Table 1. The density increased monotonically as aluminum ions were replaced by larger and heavier iron ions in both NAFP and ZAFP series. The glass transition temperature, T_g, and dilatometric softening temperature, T_s, are higher for glasses containing both iron and alumina than for glasses containing only alumina or iron for NAFP, however iron containing glasses seem to have slightly higher values. For the ZAFP glasses, T_g and T_s decreased monotonically with increasing iron content in the glass.

The thermal expansion coefficient, α, for the NAFP glasses ($> 100 \times 10^{-7}$/°C) is considerably higher than that for the ZAFP glasses ($< 70 \times 10^{-7}$/°C). It decreases as the iron content increases in the NAFP glasses, while there is no clear compositional trend for the ZAFP glasses.

The DR measured in deionized water at 90°C for 8 days for both NAFP and ZAFP glasses ranged between 10^{-7} and 10^{-9} g/cm^2 min for all the glasses investigated. The ZFP2 glass had the best aqueous chemical durability (lowest DR of 3.4×10^{-9} g/cm^2 min). For comparison, commercial soda–lime–silica glasses such as window glass, typically have a DR of $\sim 10^{-8}$ g/cm^2 min in deionized water at 70°C.

3.2. Mechanical properties of the fibers

Young's modulus, E, of the fibers is given in Table 2. The NAP3 glass fiber had the lowest modulus of 60 GPa. Young's modulus for the NAFP fibers increased from 60 GPa for the iron-free NAP3 glass to 69 GPa for the NFP3 glass which contains 20 mol% Fe_2O_3. Young's modulus of the ZAFP fibers varied little with iron content, but the ZAP2 glass fibers ($15ZnO–25Al_2O_3–60P_2O_5$, mol%) had the highest modulus at 89 GPa. Young's modulus of the starting iron phosphate glass, F40, ($40Fe_2O_3–60P_2O_5$, mol%) is also given in Table 2. This glass has the best chemical durability (smallest dissolution rate in distilled water at 90°C of $\sim 10^{-10}$ g/cm^2 min).

Weibull plots of the tensile strength for selected NAFP glass fibers are given in Figs. 2 and 3. The Weibull modulus was between 6 and 12 and as a result no error bars are given in the figures. Fig. 2 shows the tensile strength for freshly pulled fibers while Fig. 3 shows the tensile strength of fibers which had been exposed to ambient air for 10 days. The strength of the freshly pulled F40 fiber is also given in both figures for comparison. Among the freshly pulled NAFP glass fibers, the NAFP2

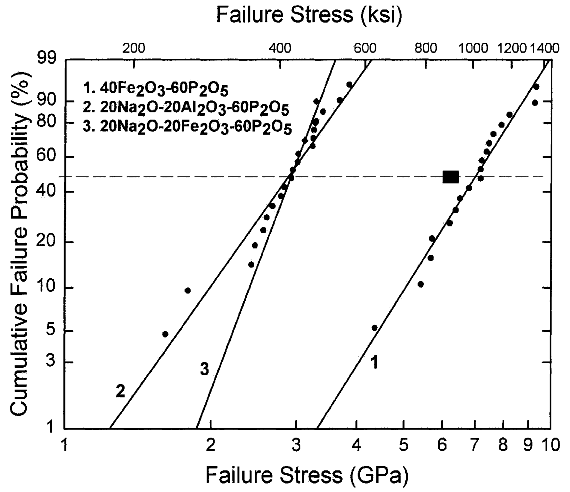

Fig. 2. Weibull probability plot of tensile strength for selected 'freshly pulled' NAFP fibers (#2-NAP3, #3-NFP3). The F40 glass (line #1) is included for comparison. The tensile strength of E-glass at 50% failure probability is indicated by the dark rectangle. Lines are drawn as a guide to the eye.

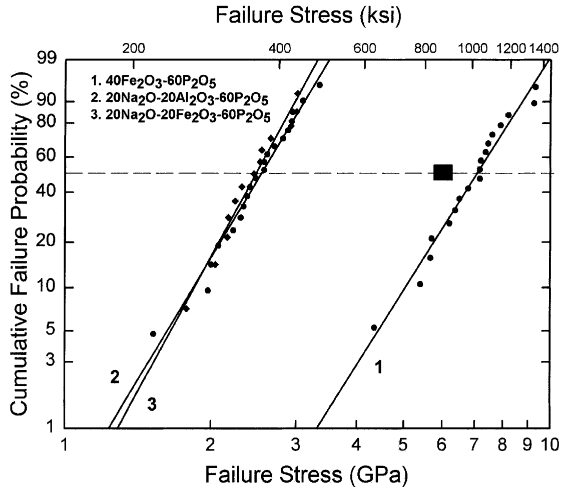

Fig. 3. Weibull probability plot of tensile strength for selected NAFP fibers after exposure to air for 10 days. The tensile strength of 'freshly pulled' F40 fiber (curve #1) and the tensile strength of E-glass at 50% failure probability (dark rectangle) is also included for comparison. Note the decrease in strength compared to 'freshly pulled' fibers in Fig. 2. Lines are drawn as a guide to the eye.

fiber had the highest strength, ~3.5 GPa at 50% failure probability (Table 2). Fibers drawn from glasses containing only alumina (NAP3) or only iron (NFP3) basically have the same failure stress of ~2.3 GPa at 50% failure probability (Fig. 2). After exposing the fibers to air for 10 days, the measured strength decreased for all the NAFP fibers (Fig. 3). The failure stress at 50% failure probability for the NAFP2 fiber decreased from 3.5 to 2.4 GPa after exposure to air.

Similarly, the Weibull plots of the tensile strength for selected freshly pulled and air exposed (10 days) ZAFP glass fibers are given in Figs. 4 and 5, respectively. Compared to NAFP fibers, ZAFP fibers were stronger as indicated by the failure stress of 7 GPa at 50% failure probability for the ZAP2 fiber. This is the highest strength for all of the fibers studied (Fig. 4, Table 2). However, a similar decrease in tensile strength occurred for the ZAFP fibers after exposure to air for 10 days (Fig. 5).

The Mössbauer spectra in Fig. 6 for NFP3 bulk glass and fiber are representative of all the other glasses and fibers. Mössbauer hyperfine parameters, isomer shift (δ) and quadrupole splitting (Δ), for all the iron containing glass and fibers are given in Table 3. The total Fe(II) fraction in the bulk glasses and fibers, see the right-hand column of Table 3, indicates that from 10% to 30% of the Fe(III) ions in the starting batch were reduced to Fe(II) during melting. A similar reduction has

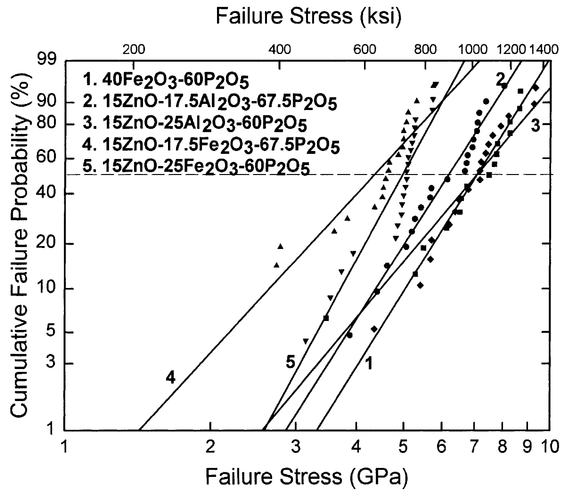

Fig. 4. Weibull probability plot of tensile strength for selected 'freshly pulled' ZAFP fibers (#2-ZAP1, #3-ZAP2, #4-NAP1, #5-NAP2). Line #1 for the F40 glass is included for comparison. Lines are drawn as a guide to the eye.

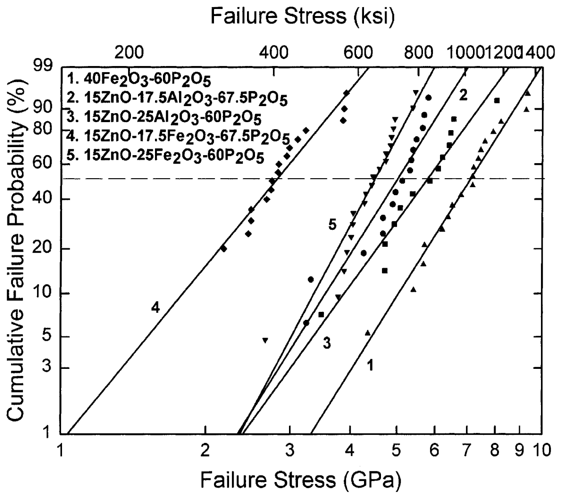

Fig. 5. Weibull probability plot of tensile strength for selected ZAFP fibers after exposure to air for 10 days. The tensile strength of 'freshly pulled' F40 fibers (line #1) is included for comparison. Lines are drawn as a guide to the eye.

been observed for many other iron phosphate glasses melted in air [4–8]. In calculating the Fe(II) fraction, the recoil-free fraction of both Fe(II) and Fe(III) was assumed to be the same.

The isomer shift values are typical of both Fe(II) and Fe(III) ions in octahedral co-ordination in the bulk glass and corresponding fiber. The Fe(II) fraction in the bulk glass and fiber is essentially the same, but it varies among the different glass compositions. The NFP3 glass has the lowest amount of Fe(II) fraction at 7% while the ZAFP2 glass has the highest Fe(II) fraction at 34%.

The IR spectra of selected NAFP and ZAFP bulk glass and corresponding fiber are given in Fig. 7. The IR spectra of the fibers are shown as dashed lines below the corresponding bulk glass spectra. There are four absorption bands between 450 and 1300 cm^{-1} in the IR spectra of the NAFP and ZAFP glass and fibers. The bands at 1280 and 1150 cm^{-1} have been assigned to the asymmetric and symmetric stretching vibrations of the two non-bridging oxygen atoms bonded to phosphorus atoms in PO_2 units (O–P–O) [14,15]. The absorption bands at 910, 760 and 490 cm^{-1} are assigned to asymmetric stretching, symmetric stretching, and bending vibrations of P–O–P bonds, respectively [14,15]. The band at 520 cm^{-1} is observed only in

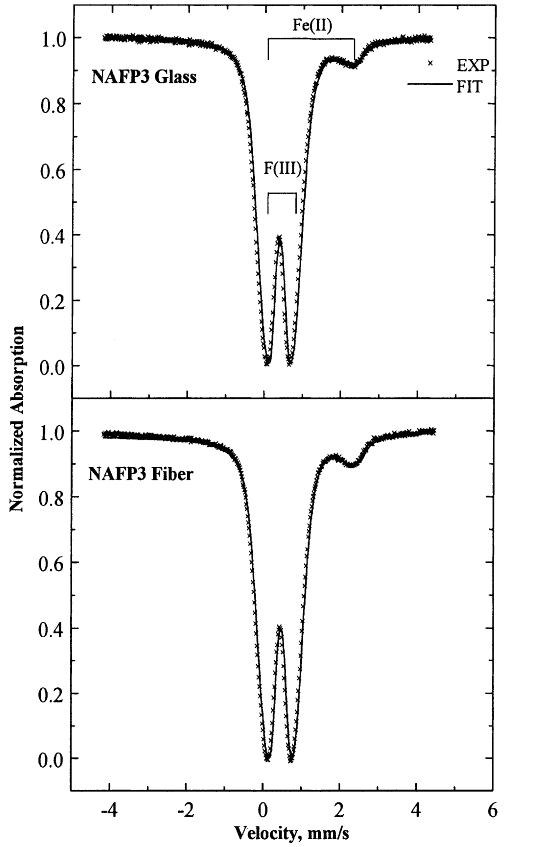

Fig. 6. Room temperature Mössbauer spectra and fits for NAPF3 bulk glass and fiber. These spectra are representative for the other glasses.

the NFP3 ($20Na_2O$–$20Fe_2O_3$–$60P_2O_5$, mol%) glass and fiber spectra. There are small differences in the IR spectra of the other glasses, but there is no appreciable difference in the spectra of the glass and corresponding fiber indicating that the structure of the bulk glass and fiber are very similar.

4. Discussion

4.1. General properties of bulk glasses

The density of bulk glasses ranged from a low of 2.61 g/cm^3 for NAP3 to a high of 3.39 g/cm^3 for ZFP1. The addition of iron oxide to the glass increases the density in both series of glasses. The glass transition temperature also increased with

Table 3
Room temperature Mössbauer hyperfine parameters, isomer shift (δ) and quadrupole splitting (Δ), and fraction of ferrous ions, Fe(II), in both bulk glass (B) and fibers (F)

Sample	δ (mm/s)		Δ (mm/s)		Fe(II) fraction
	Fe(II)	Fe(III)	Fe(II)	Fe(III)	
F40B	1.20	0.38	2.14	0.90	0.21
F40F	1.21	0.37	2.07	0.89	0.18
NAFP1B	1.21	0.46	2.18	0.60	0.19
NAFP1F	1.23	0.44	2.17	0.73	0.16
NAFP2B	1.25	0.44	2.13	0.70	0.11
NAFP2F	1.20	0.44	2.17	0.73	0.15
NAFP3B	1.27	0.43	2.00	0.84	0.08
NAFP3F	1.17	0.44	2.26	0.74	0.10
NFP3B	1.19	0.44	2.21	0.77	0.07
NFP3F	1.17	0.44	2.19	0.79	0.07
ZFP1B	1.23	0.41	2.15	0.85	0.18
ZFP1F	1.23	0.40	2.10	0.86	0.16
ZFP2B	1.20	0.40	2.22	0.87	0.20
ZFP2F	1.20	0.39	2.15	0.89	0.17
ZAFP1B	1.26	0.42	2.05	0.77	0.32
ZAFP1F	1.24	0.43	2.08	0.77	0.31
ZAFP2B	1.22	0.43	2.13	0.75	0.34
ZAFP2F	1.23	0.43	2.10	0.79	0.32

The estimated error in these parameters is ±0.03 mm/s.

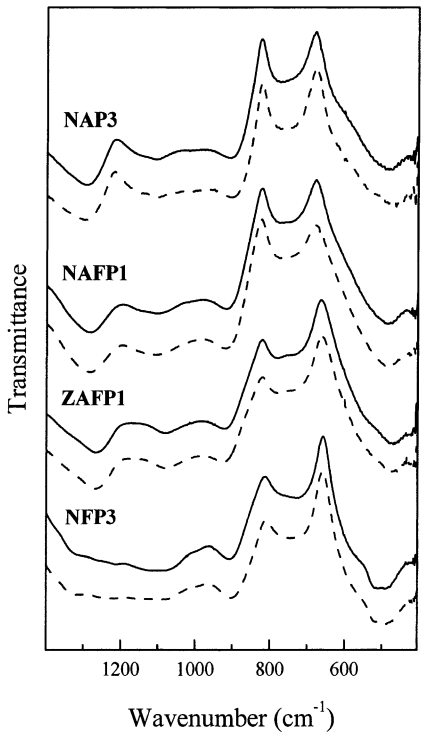

Fig. 7. IR spectra of selected bulk glasses (solid lines) and fibers (dashed lines).

the addition of iron to the NAFP glasses while T_g decreased with the addition of iron to the ZAFP glasses.

The thermal expansion coefficient for the alkali containing NAFP glasses is considerably higher than that for the ZAFP glasses. The addition of alkali ions in a phosphate glass is known to decrease the connectivity of the glass structure by creating non-bridging oxygens, hence increasing the thermal expansion coefficient. This is reflected in a higher thermal expansion coefficient for the NAFP glasses compared to the alkali-free ZAFP glasses. The addition of aluminum and iron oxide does not seem to have noticeable effect on the thermal expansion coefficient.

The dissolution rates of the NAFP and ZAFP glasses given in Table 1 are much smaller than those of binary alkali phosphate glasses. The addition of iron or aluminum oxide to phosphate glasses is known to increase the chemical durability (lower dissolution rate). In iron phosphate glasses, the increase in chemical durability has been attributed to the formation of Fe–O–P type bonds in the glass structure at the expense of the more easily hydrated P–O–P bonds [4–8].

4.2. Mechanical and structural properties

The tensile strength and Young's modulus of the glasses in this work are compared with those for silica and other phosphate glasses reported previously in Table 2. Vitreous silica fiber has a tensile strength in liquid nitrogen of 14 GPa which is the highest among the oxide glasses. It is believed that the three-dimensional network consisting of strong Si–O–Si bonds is responsible for such a high tensile strength.

In general, the tensile strength of the phosphate glass fibers in the present work falls into two categories. The ZAFP glass fibers have a tensile strength in the 4.2–7.2 GPa range while the tensile strength of the NAFP fibers is in the 2.8–4.2 GPa range. The tensile strength of the F40 glass fiber is also 7.2 GPa, the same as that for the ZAP2 fiber. Although not as clear, Young's moduli of the ZAFP glasses appear to be higher than those for the NAFP glasses. A similar increase in tensile strength was observed [15] for a $Zn(PO_3)_2$ glass (2.4 GPa) compared to that for a $NaPO_3$ glass (1.8 GPa). This increase in strength is believed to be due to the replacement of monovalent sodium ions by divalent zinc ions which increases the cross-link density in the glass structure and makes the glass stronger.

Replacing Al_2O_3 by Fe_2O_3 in the sodium aluminophosphate glasses (NAP) does not appear to have any discernable effect on the strength. However, zinc aluminophosphate (ZAP) glasses have a higher tensile strength than the zinc iron phosphate (ZFP) glasses. We note that iron containing (ZFP) glasses have 7–20% Fe(II) in the glass (Table 3). Thus, the existence of Fe(II) in the ZFP glasses may be the reason for the lower observed strength due to the lower field strength of ferrous ions compared to ferric ions. However, the F40 glass also contains 20% Fe(II), but its strength is 7.2 GPa. It is clear, however, that replacing monovalent sodium ions by divalent zinc ions in the glass composition has the largest effect on the measured tensile strength.

It has been shown that the main factor determining the strength of glasses is the degree to which their anion network is bonded [16]. The structure of vitreous silica consists of corner shared SiO_4 tetrahedra with four bridging oxygen ions bonded to neighboring SiO_4 tetrahedra while the structure of phosphate glasses consists of PO_4 tetrahedra with three bridging oxygen ions bonded to neighboring tetrahedral units. Hence, the strength of silica fiber is expected to be high among glasses, and this is in fact observed (14 GPa, Table 2).

The addition of alkali ions to the phosphate network depolymerizes the phosphate network and decreases the connectivity. Aluminum can be incorporated into phosphate structure as a network former as AlO_4 units thus contributing to the anion network bonding by converting the P=O bonds into bridging oxygens. The local charge balance is maintained by the alkali ions. Hence, it is expected that adding alumina to sodium phosphate glasses would increase the strength of the glass. A similar increase was observed in Na_2O–SiO_2 glasses with the addition of Al_2O_3 [16]. Zn^{2+} ions can also be in four fold co-ordination in phosphate glasses and, as mentioned above, can increase the cross-linking in the glass structure, thereby, increasing the strength [17].

Young's modulus and tensile strength of the phosphate glasses investigated in the present work are comparable to or higher than those of commercially produced, better known silicate glass fibers (Table 2). Young's modulus of the ZAP2 glass fiber (89 GPa) is essentially the same as that of S-glass (87 GPa). Both ZAP2 and F40 glass fibers have a higher tensile strength (7.2 GPa) than E-glass (5.4 GPa) and are only slightly weaker than S-glass (8.4 GPa). The combination of high strength and good chemical durability of the iron phosphate glasses, along with the fact that they can be prepared at relatively low temperatures, are valuable advantages for potential technological applications.

The tensile strength of all the glass fibers investigated in the present study decreases after exposure to air for 10 days. This decrease ranges from 15% to 34% of the strength of 'freshly pulled' fibers. Preliminary investigation of the surface morphology of the freshly pulled and air exposed fibers for the F40 iron phosphate glass was done using scanning electron microscopy (SEM), see Figs. 8(a) and (b). The surface of the freshly pulled

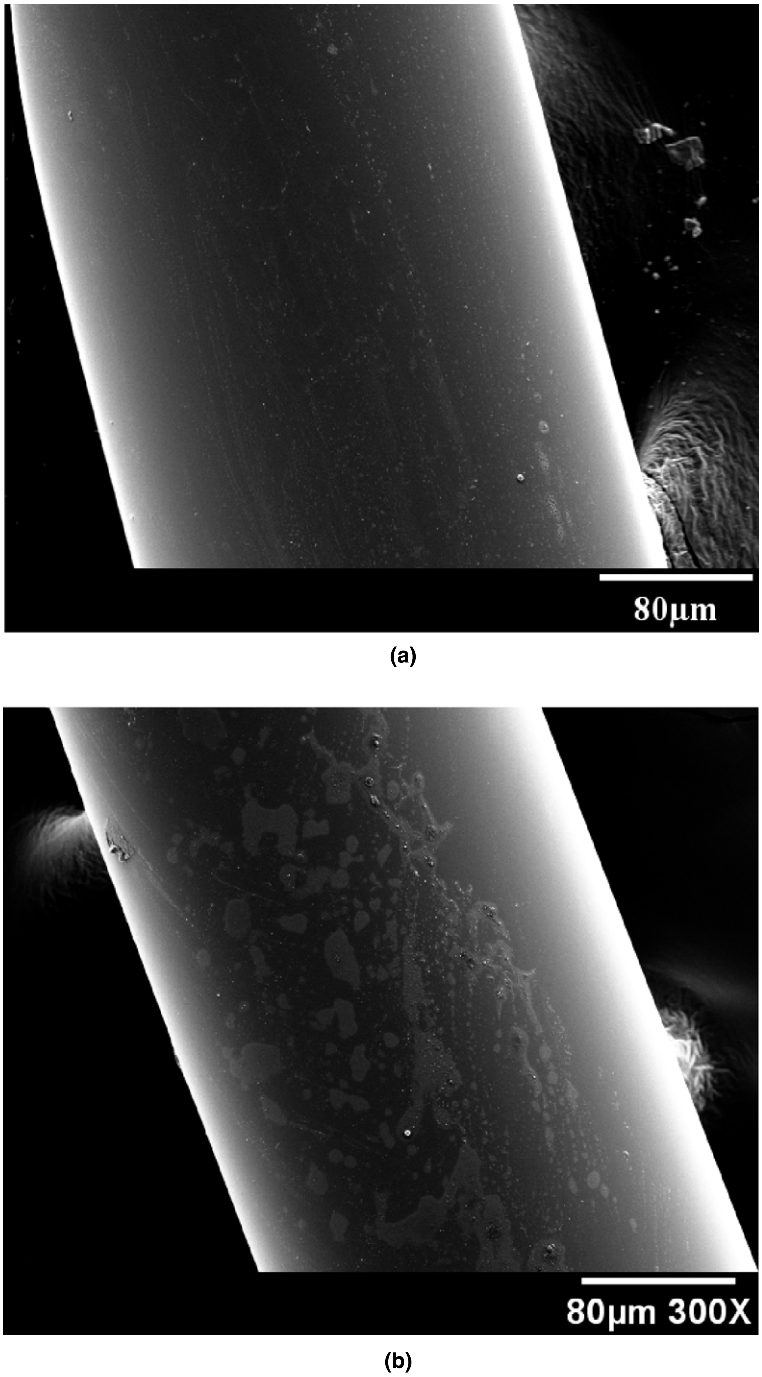

Fig. 8. (a) An SEM image of 'freshly pulled' F40 glass fiber. (b) An SEM image of an F40 fiber exposed to air at room temperature for 30 days. Note that the surface of the 'freshly pulled' fiber is smooth compared to the surface of the fiber exposed to air.

F40 fiber was fairly smooth (Fig. 8(a)) while the surface of the air-exposed fibers (Fig. 8(b)) was rough and exhibited spots over the surface. This indicates that a reaction on the surface is occurring, most likely with moisture which ultimately causes the observed decrease in the strength. This reaction with moisture is continuing to be studied.

Mössbauer hyperfine parameters obtained from iron containing glass and fibers indicate that both Fe(II) and Fe(III) ions are octahedrally co-ordinated to nearby oxygen ions. The Fe(III) ions are reduced more in the ZAFP1 and ZAFP2 glasses compared to the other glasses studied.

The overall PO_4 network in bulk glass and fiber remains basically the same as indicated by the IR spectra (Fig. 7). There is no difference between the IR spectra of the bulk glass and corresponding fiber. It appears that the band at 1150 cm^{-1}, assigned to the asymmetric stretching mode of PO_2 units, is absent in the IR spectra of the NFP3 glass and fiber. Another band at $\sim$ 1350 cm^{-1} appears to emerge in the IR spectrum of the NFP3 glass which is attributed to the stretching mode of P=O double bonds [14].

5. Conclusion

The mechanical and structural properties of sodium and zinc iron–aluminum–phosphate fibers have been investigated. Young's moduli of the fibers were in the 60–89 GPa range. The tensile strength of the ZAFP fibers (4.2–7.2 GPa) was higher than the tensile strength of the NAFP fibers (2.8–4.2 GPa). After exposing the fibers to air for 10-days, the strength of all fibers decreased from 15% to 35%. Mössbauer and IR spectroscopy indicated that the structure of bulk glass and corresponding fiber was very similar for the compositions studied.

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References

[1] R.K. Brow, J. Non-Cryst. Solids 263&264 (2000) 1.
[2] B.C. Sales, L.A. Boatner, Science 226 (1984) 45.
[3] D.E. Day, Z. Wu, C.S. Ray, P. Hrma, J. Non-Cryst. Solids 241 (1998) 1.
[4] G.K. Marasinghe, M. Karabulut, C.S. Ray, D.E. Day, M.G. Shumsky, W.B. Yellon, C.H. Booth, P.G. Allen, D.K. Shuh, J. Non-Cryst. Solids 222 (1997) 144.
[5] M. Karabulut, G.K. Marasinghe, C.S. Ray, D.E. Day, O. Ozturk, G.D. Waddill, J. Non-Cryst. Solids 249 (1999) 106.
[6] C.H. Booth, P.G. Allen, J.J. Bucher, N.M. Edelstein, D.K. Shuh, G.K. Marasinghe, M Karabulut, C.S. Ray, D.E. Day, J. Mater. Res. 14 (1999) 2628.
[7] M. Karabulut, G.K. Marasinghe, C.S. Ray, G.D. Waddill, D.E. Day, Y.S. Badyal, M.-L. Saboungi, S. Shastri, D. Haeffner, J. Appl. Phys. 87 (2000) 2185.
[8] C.S. Ray, X. Fang, M. Karabulut, G.K. Marasinghe, D.E. Day, J. Non-Cryst. Solids 249 (1999) 1.
[9] T. Klug, R. Bruckner, J. Mater. Sci. 29 (1994) 4013.
[10] A. Rouzaud, E. Barbier, J. Ernoult, E. Quesnel, Thin Solid Films 270 (1995) 270.
[11] M.J. Matthewson, C.R. Kurkjian, S.T. Gulati, J. Am. Ceram. Soc. 69 (1986) 815.
[12] M.J. Matthewson, C.R. Kurkjian, J. Am. Ceram. Soc. 70 (1987) 662.
[13] M.D. Dyar, Am. Miner. 70 (1985) 304.
[14] J.J. Hudgens, S.W. Martin, J. Am. Ceram. Soc. 76 (1993) 1691.
[15] C.R. Kurkjian, J. Non-Cryst. Solids 263&264 (2000) 207.
[16] T.I. Pesina, L.G. Baikova, V.P. Pukh, I.I. Novak, M.F. Kireenko, Sov. Phys. Chem. Glasses 12 (1986) 14.
[17] L.G. Baikova, Y.K. Fedorov, M.N. Tolstoi, V.P. Pukh, L.V. Tikhonova, S.G. Lunter, A.B. Sinani, Sov. Phys. Chem. Glasses 16 (1989) 211.